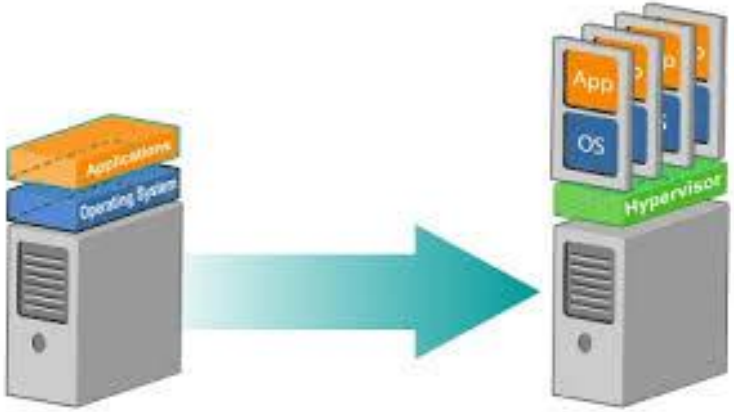
Virtualization and Cloud Computing



By,

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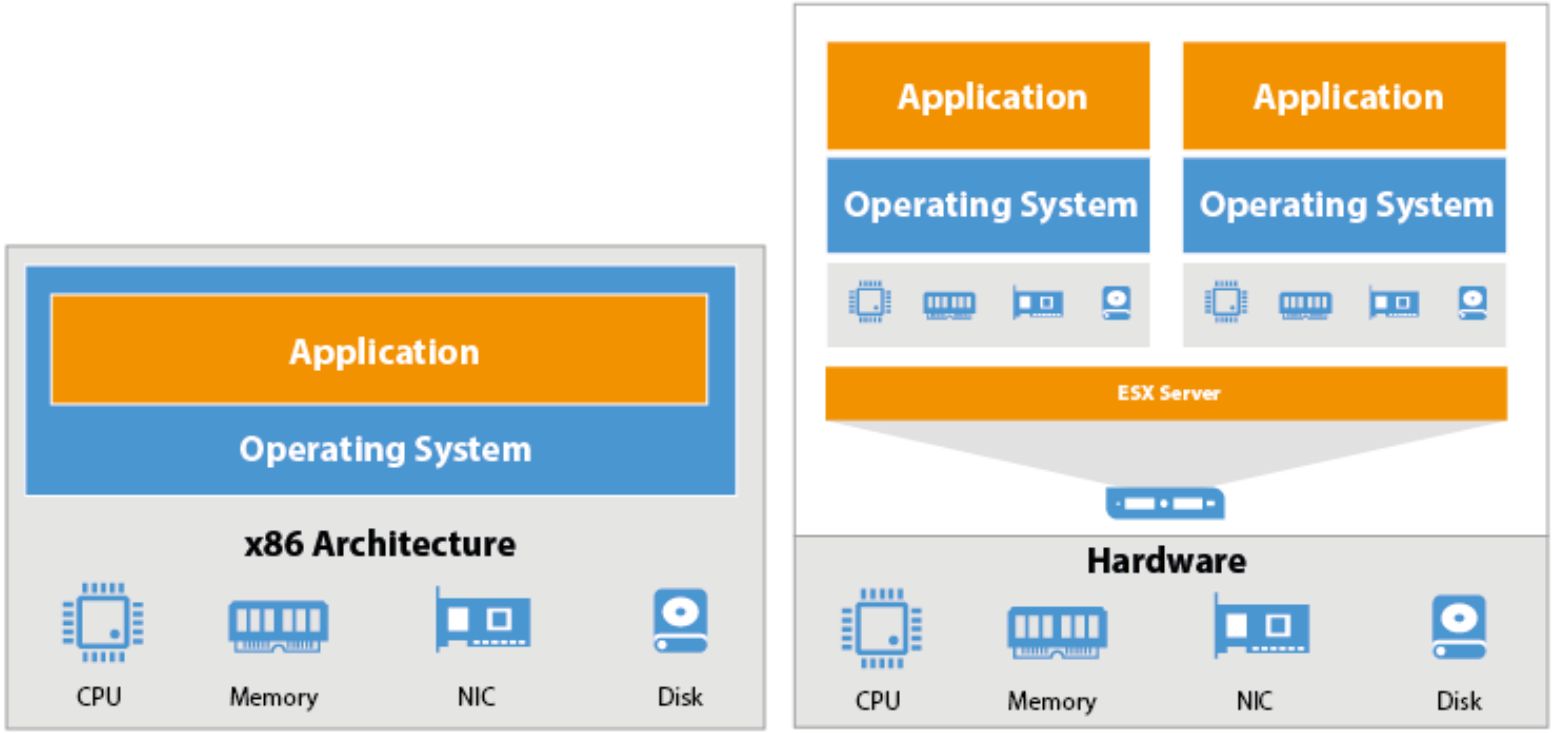
Traditional Server vs Virtualization



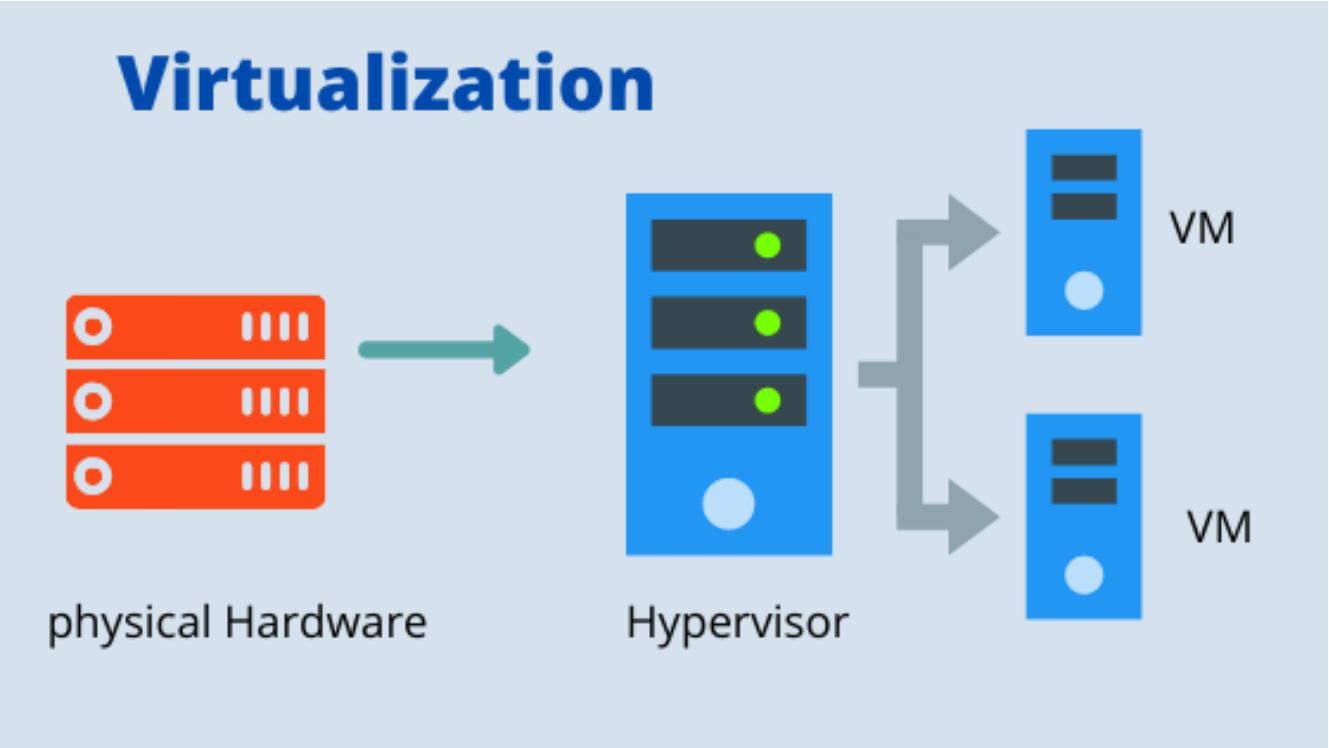
Traditional Physical Servers: Characteristics

- **Dedicated Hardware:** Each server runs on its own dedicated hardware, including CPU, memory, storage, and network resources.
- **Single Operating System:** Each physical server typically runs a single operating system (OS) instance.
- **Resource Utilization:** Often results in underutilization of resources since dedicated servers may not fully leverage their hardware capabilities.
- **Isolation:** Physical isolation provides strong security and stability because issues in one server don’t affect other

Virtualization: Characteristics

- **Shared Hardware:** Multiple virtual machines (VMs) run on a single physical server, sharing its resources.
- **Multiple Operating Systems:** Each VM can run a different OS instance, even different OS types.
- **Resource Utilization:** Improved resource utilization since multiple VMs can leverage the same hardware.
- **Isolation:** Software-based isolation between VMs provides separation but can be less robust than physical isolation.

Virtualization : What It is ?



Virtualization : What It is ?

- Virtualization is a technology that transforms hardware into software.

- Virtualization is a technology that enables the creation of virtual instances of computing resources, such as hardware platforms, storage devices, and network resources, within a single physical system.

- It abstracts the physical characteristics of these resources from their users, making it possible to run multiple independent virtual environments simultaneously on the same physical hardware.

Virtualization : Benefits ??

- **Improved Resource Utilization:** Maximizes the use of physical hardware by running multiple virtual machines (VMs) on a single physical machine.
- **Cost Savings:** Reduces the need for physical hardware, lowering capital and operational expenses.
- **Scalability and Flexibility:** Easily scales resources up or down based on demand and quickly adapts to changing workloads.
- **Enhanced Disaster Recovery:** Simplifies backup and recovery processes, ensuring business continuity.
- **Isolation and Security:** Provides isolated environments for applications and systems, enhancing security and reducing the risk of conflicts.

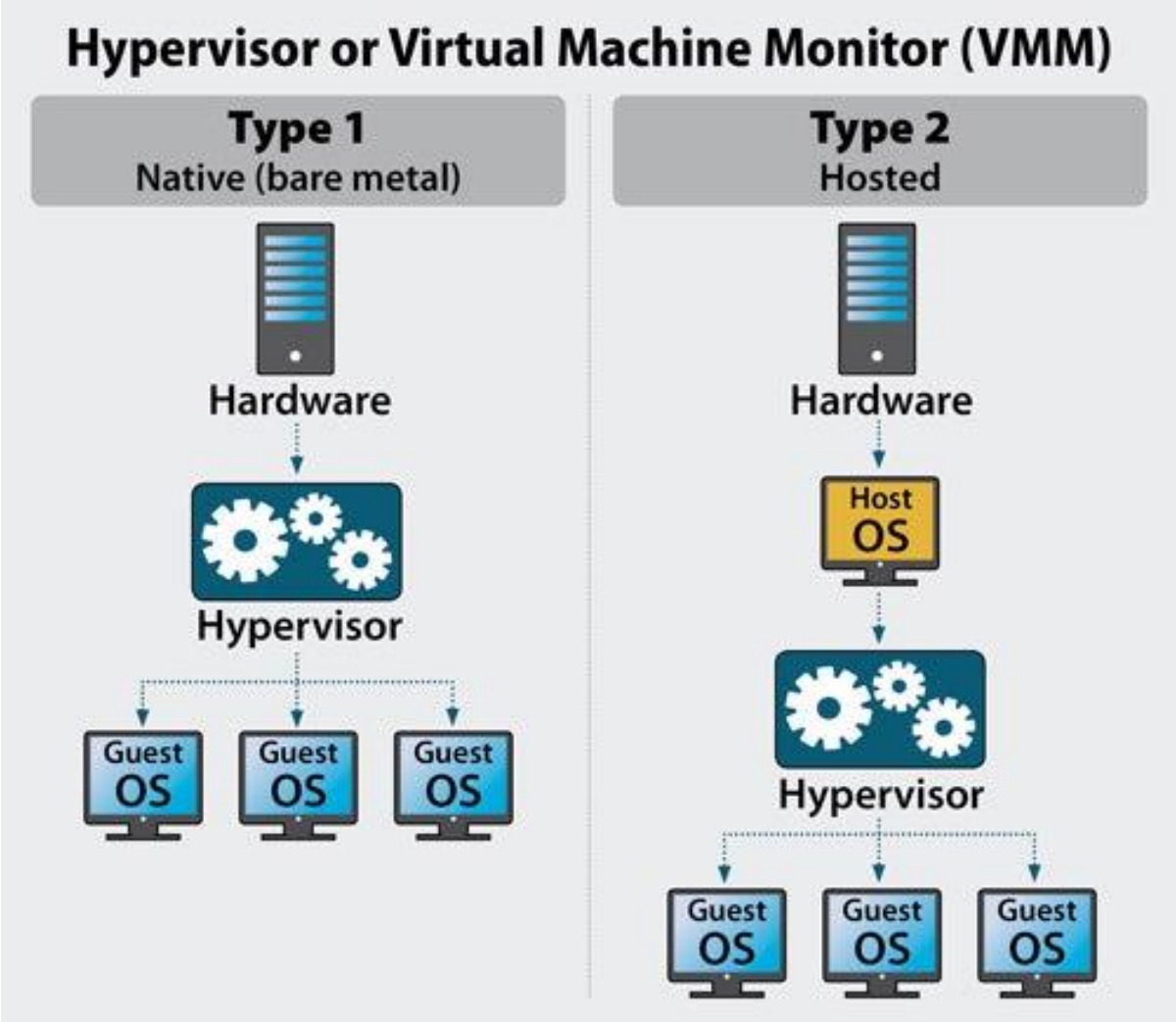
Hypervisor

The hypervisor is the virtualization software that you install on your physical machine. It is a software layer that acts as an intermediary between the virtual machines and the underlying hardware or host operating system. The hypervisor coordinates access to the physical environment so that several virtual machines have access to their own share of physical resources. The following are the two main types of hypervisors.

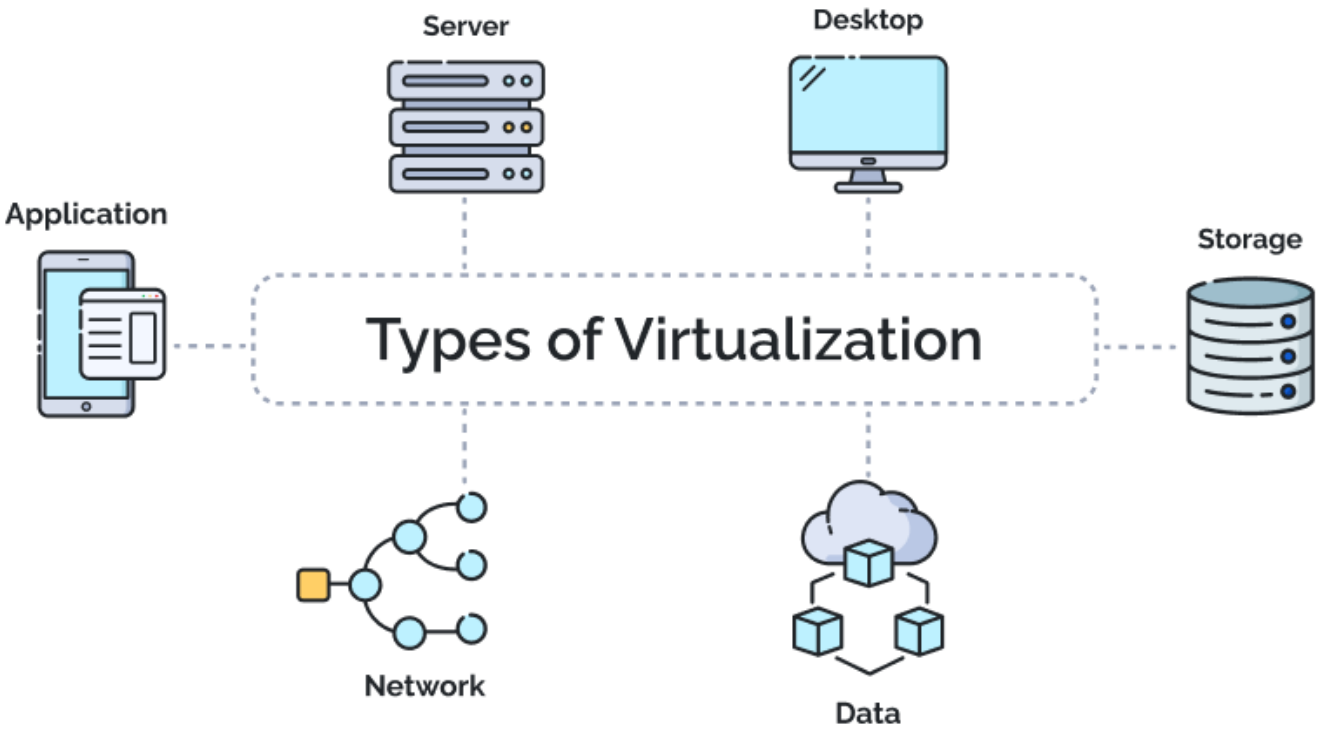
Type 1 hypervisors
A type 1 hypervisor—also called a bare-metal hypervisor—runs directly on the computer hardware. It has some operating system capabilities and is highly efficient because it interacts directly with the physical resources. **(ESXi, Hyper-V, Xen)**

Type 2 hypervisors
A type 2 hypervisor runs as an application on computer hardware with an existing operating system. Use this type of hypervisor when running multiple operating systems on a single machine. **(VMWare Workstation, Oracle Virtual Box)**

Hypervisor


Hypervisor or Virtual Machine Monitor (VMM)
Type 1
Native (bare metal)
Hardware
Hypervisor
Guest OS
Guest OS
Guest OS
Type 2
Hosted
Hardware
Host OS
Hypervisor
Guest OS
Guest OS
Guest OS

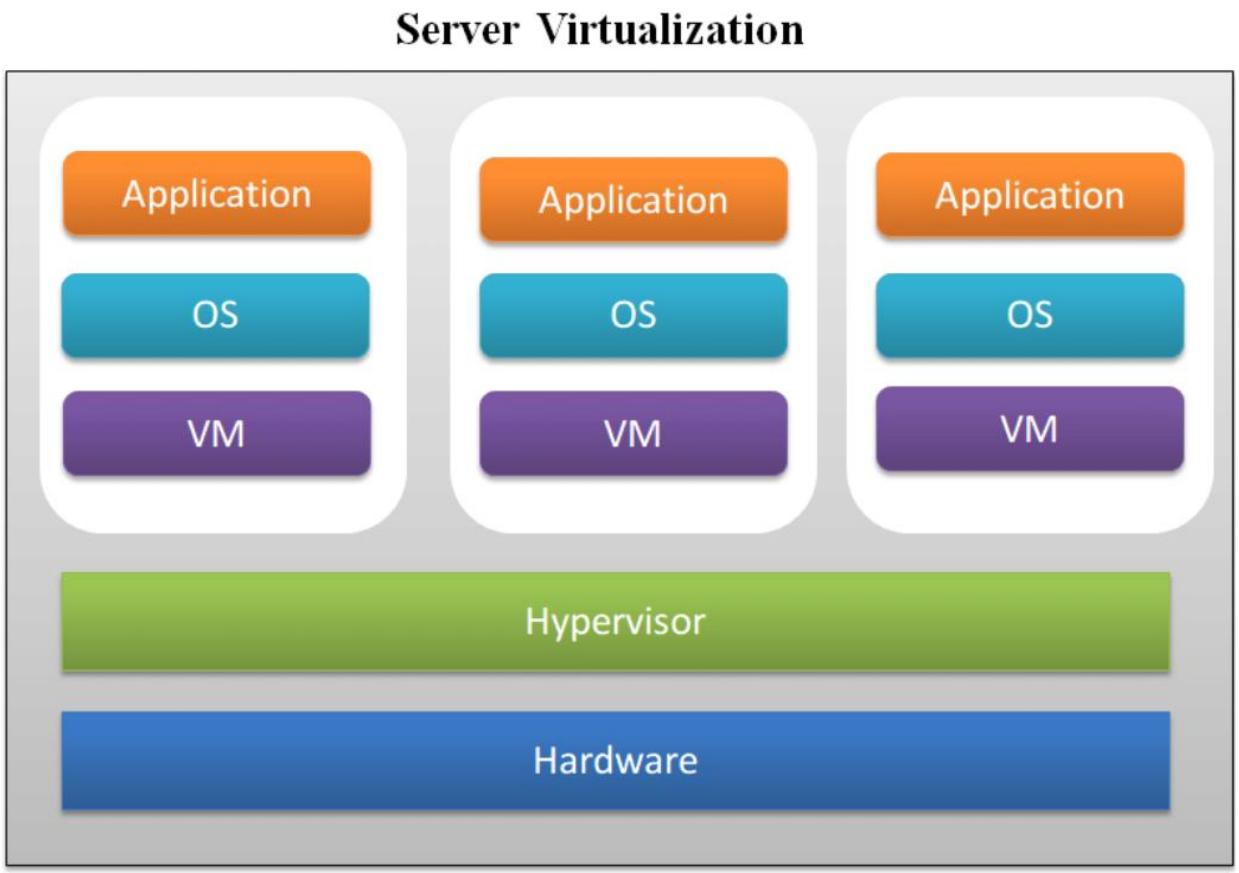
Virtualization : Types



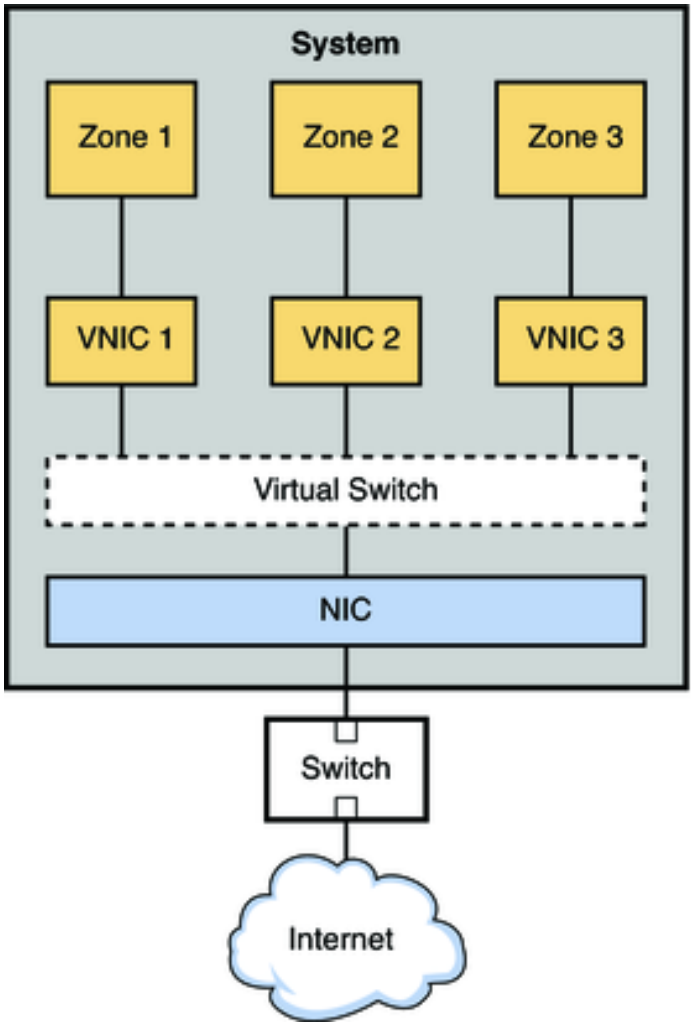
Virtualization : Types

- **Server Virtualization**: Divides a physical server into multiple virtual servers, each running its own operating system and applications. This enhances resource utilization and reduces costs.
- **Storage Virtualization**: Combines multiple physical storage devices into a single, centralized storage resource that can be managed more efficiently.
- **Network Virtualization**: Creates multiple virtual networks within a single physical network infrastructure, allowing for more flexible and efficient network management.
- **Desktop Virtualization**: Runs desktop operating systems and applications on a centralized server, enabling access from any device, improving manageability and security.
- **Application Virtualization**: Separates applications from the underlying hardware and operating system, allowing them to run in isolated environments, which simplifies deployment and increases compatibility.

Virtualization : Server Virtualization

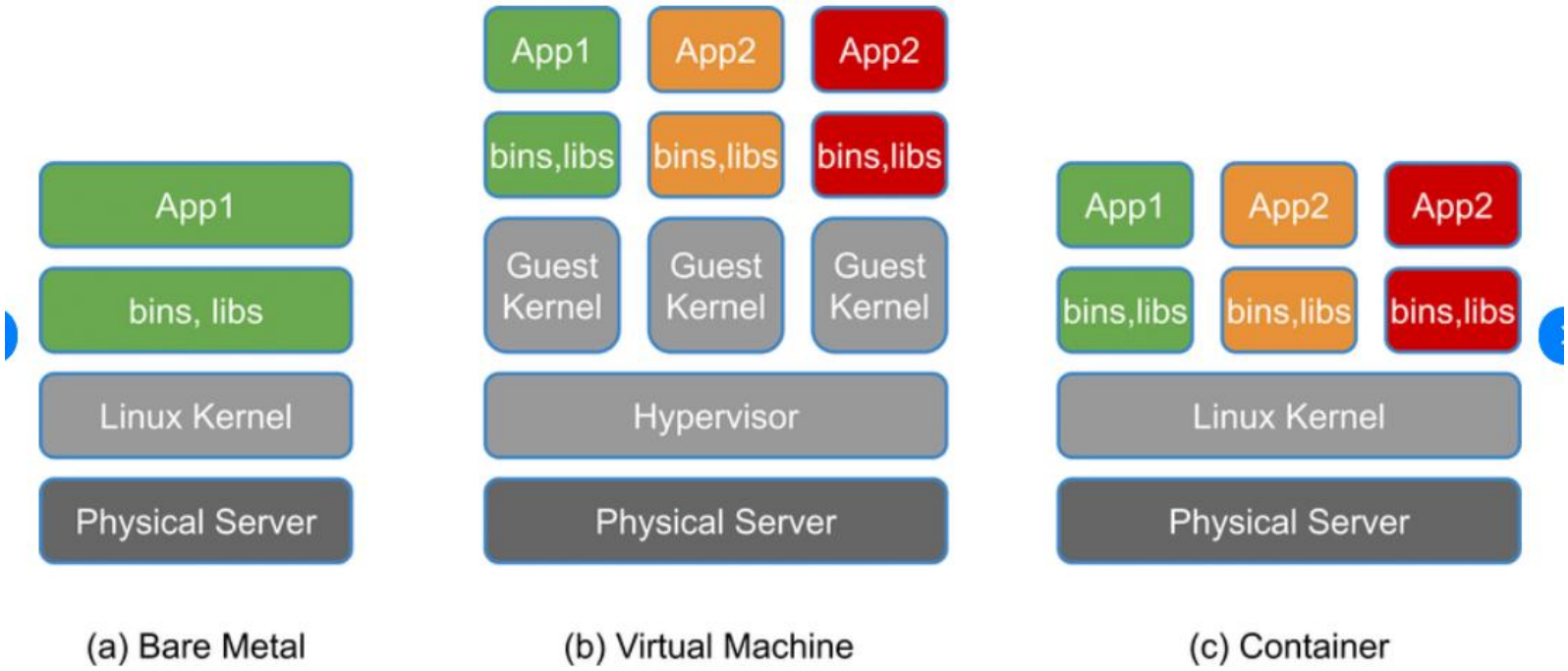


Virtualization : Network Virtualization



Virtualization : Desktop Virtualization

Bare Metal vs Virtualization vs Containerization



Virtualization Platforms

- **VMware vSphere**: Best for large enterprises needing robust features and high availability.
- **Microsoft Hyper-V**: Ideal for organizations with a strong Microsoft ecosystem.
- **KVM**: Suitable for Linux environments and open-source projects.
- **Citrix Hypervisor**: Great for VDI and cloud environments needing advanced GPU capabilities.
- **Oracle VM VirtualBox**: Perfect for individual developers and small-scale virtualization.
- **Proxmox VE**: An excellent choice for SMBs and open-source users looking for a hybrid approach.

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Top Virtualization Software in 2024

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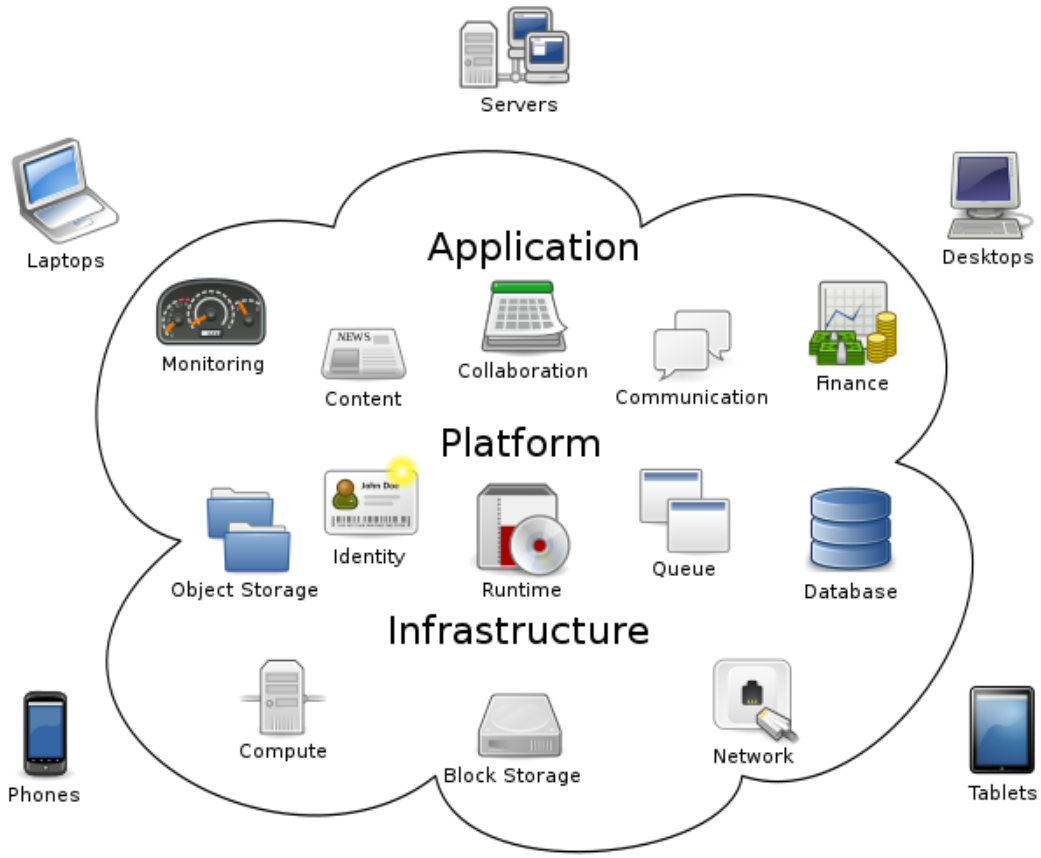
Cloud Computing



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Assignment

Discuss the Importance of Virtualization and Cloud Computing in today's Business Model with a Suitable Example.

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Thank You